

Marcos Guerrero Villamor

marcosguerrerov27@gmail.com | +34 611 408 310 | mguerrerov.netlify.app | GitHub | LinkedIn
Toledo, Spain — EU work authorisation

PROFESSIONAL PROFILE

Data and **Machine Learning** profile with a web development background and over a year of experience in corporate **IT support**. Built batch pipelines with **PySpark** and **Parquet** (1.31 billion events processed) and classification models with **Scikit-Learn** and **PyTorch**. **AWS Certified Cloud Practitioner**.

EDUCATION

- Specialisation Course in Artificial Intelligence and Big Data (EQF 5)** 2025 – 2026
IES Azarquiel *Toledo, Spain*
◦ **Technologies:** Python, Scikit-Learn, PyTorch, Pandas, NumPy, Spark, Power BI
- Higher Technician in Web Application Development (EQF 5)** 2023 – 2025
IES Julio Verne *Toledo, Spain*
◦ **Technologies:** Python, PHP, JavaScript, MySQL, PostgreSQL
- Technician in Computer Systems and Networks (EQF 4)** 2021 – 2023
IES Julio Verne *Toledo, Spain*
◦ **Technologies:** Windows, Linux, networking and end-user support

PROFESSIONAL EXPERIENCE

- Irium Soluciones y Sistemas** Sep 2025 – Oct 2025
IT Support Technician *Toledo, Spain*
◦ Diagnosed and resolved **hardware, software and network** incidents in a corporate environment.
◦ Configured and deployed workstations, providing remote and on-site support through a **ticketing** system.
- Cibersia Informática Toledo** Mar 2025 – Jun 2025
Web Developer *Olías del Rey, Spain*
◦ Web development with **PHP**, **WordPress** and **Prestashop**: custom modules and **MySQL** database maintenance.
◦ Delivered **performance and security** improvements on live production sites.
- Fibritel Telecomunicaciones SLU** Mar 2023 – Jun 2024
IT Support Technician *Illescas, Spain*
◦ Resolved hardware, software and connectivity incidents; preventive and corrective systems maintenance.

TECHNICAL PROJECTS

- GH Archive Pipeline**
Batch ingestion and Bronze-Silver model with PySpark
◦ Processed **1.31 billion events** (12.5 months of public GitHub activity) downloaded hour by hour and converted into queryable **Parquet**: 149.4 GiB on disk, 0.54× compression against the source.
◦ Built **download, bronze, silver and quality** layers with deduplication: 0.0009 % residual duplicates.
◦ Runs end to end on a local desktop machine at **€0 infrastructure cost**; tested with **pytest**, CI on **GitHub Actions**.
- Candidate Recommender**
Semantic ranking of CVs against a job posting
◦ Ranked candidates by semantic proximity using **384-dimension embeddings** instead of keyword matching, with a re-ranking step that penalises unmet hard requirements.
◦ **FastAPI** backend and **Streamlit** interface over **MySQL**; **61-test** suite running in 6 seconds.
- Credit Card Fraud Detection**
Classification on heavily imbalanced data
◦ Benchmarked three class-imbalance strategies (incl. **SMOTE**) across two models on a **284,807-transaction** dataset with 0.17 % fraud rate.
◦ Best result: **0.8364 F1** and **98.57 % precision** with **Random Forest** and no resampling.

TECHNICAL SKILLS

Languages: Python, SQL, PHP, JavaScript, HTML, CSS
Data & Big Data: PySpark, Parquet, Pandas, NumPy, Power BI, embeddings
Modelling: Scikit-Learn, PyTorch, Random Forest, SMOTE, validation and metrics
Tools & Cloud: FastAPI, Streamlit, Git, GitHub Actions, pytest, n8n, Claude Code, AWS
Databases & Systems: MySQL, PostgreSQL, Linux, Windows, networking and ticketing

CERTIFICATIONS & ADDITIONAL

- **AWS Certified Cloud Practitioner** — Amazon Web Services
- **Languages:** Spanish (Native), English (B2-C1)
- **Availability:** Flexible shifts, driving licence, own vehicle — Portfolio